

Solar Energy Production Forecasting with ARIMAX

IE360 Time Series Analysis Term Project Report

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Abstract

This report presents a time-series forecasting pipeline for predicting next-day hourly solar energy production. The project was developed as the IE360 Time Series Analysis term project and ranked 2nd out of 30 groups in the project performance evaluation. The final implementation combines historical solar production, Open-Meteo weather forecasts, calendar variables, lagged production features, and an ARIMAX-style SARIMAX model with exogenous regressors. The model first forecasts total daily solar production and then distributes the daily total into 24 hourly values using historical intra-day production profiles. A local backtest around the last available production date produced a daily WMAPE of 7.44% and an hourly WMAPE of 9.28% over the recent evaluation window.

1 Introduction

Solar production forecasting is a practical time-series problem because the observed production series depends on both temporal structure and exogenous weather conditions. Solar output has a strong daily shape, seasonal variation, and direct sensitivity to radiation and cloud cover. The objective of this project is to forecast the next day's hourly solar production, denoted as `sun_rt`, and print the required result as a Python list of 24 values.

The final project code is implemented in `forecast.py`. The script fetches production and weather data at runtime, constructs a leakage-aware feature table, trains a SARIMAX model with exogenous variables, and returns the next-day hourly forecast. The current cleaned version also separates the forecasting pipeline into readable functions, making the project easier to inspect, validate, and extend.

2 Problem Definition and Output Constraint

The target variable is hourly solar production:

$$y_t = \text{sun_rt}_t.$$

For each execution date, the model must generate one prediction for each hour of the next day:

$$\hat{y}_{d+1,0}, \hat{y}_{d+1,1}, \dots, \hat{y}_{d+1,23}.$$

The output constraint is operationally simple but strict: the final printed line of the script must be a Python list containing exactly 24 numeric values. The model therefore needs to handle missing recent production data, future weather rows, and possible API delays without breaking the final output format.

3 Data Sources

3.1 Production Data

The dependent variable is loaded from a Google Drive CSV file through `get_production_data()` in `forecast_helper.py`. The table contains timestamped observations of `sun_rt`. Timestamps are converted to the `Europe/Istanbul` timezone before merging with weather data.

Table 1 summarizes the local report run. The production data currently ends on 2025-05-27, while the report was generated on a later execution date. For that reason, current next-day forecasts cannot be evaluated against actual values yet; validation is performed by anchoring the pipeline around the last observed production date.

Table 1: Data coverage from the local report run.

Metric	Value
Hourly rows in constructed table	38,712
Observed hourly production rows	29,832
Production date range	2022-01-01 to 2025-05-27
Forecast table end date	2026-06-01
Daily rows after aggregation	1,613
Final model feature count	16

3.2 Weather Data

Weather data is downloaded from Open-Meteo for two plant coordinates. The model uses the following hourly weather variables:

- temperature at 2 meters,
- shortwave radiation,
- cloud cover,
- relative humidity,
- weather code.

The script combines historical forecast data and current forecast data. When timestamps overlap, historical rows are prioritized because they are treated as the best available past forecast history for training.

4 Preprocessing and Feature Engineering

The model constructs a continuous hourly time index from 2022-01-01 through the forecast day. Production data and weather data are merged into this timeline by timestamp. A 3-day lag is then created:

$$\text{sun_rt_lag_3days}_t = \text{sun_rt}_{t-72}.$$

This lag is used because the latest production observations may not be available at the time a next-day forecast is required. Using a 72-hour shift is a simple way to keep the model closer to a realistic operational setting.

Weather features from the two plant locations are summarized into location-level aggregates. The most important derived weather feature is effective radiation:

$$\text{effective_radiation} = \text{shortwave_radiation_mean} \times \left(1 - \frac{\text{cloudcover_mean}}{100}\right).$$

Hourly rows are then aggregated into a daily modeling table. Table 2 shows the feature blocks in the final model.

Table 2: Feature blocks used in the final SARIMAX model.

Feature block	Number of features
Weather and transformed weather variables	7
Lagged production	1
Day-of-week dummy variables	7
Trend index	1
Total	16

All continuous production and weather features used by the model are transformed with $\log(1+x)$. Negative values are clipped to zero before transformation. This keeps zero-production periods valid and reduces the effect of very large daily production totals.

5 Missing Values

Missing values are expected because production data ends before the live weather forecast horizon, and future target values are naturally unavailable. Table 3 reports missingness from the local report run.

Table 3: Missing-value summary from the local report run.

Area	Missing rate
Hourly production target	22.94%
Hourly weather raw columns	0.00%
Daily feature matrix	0.00%
Next-day forecast rows	9.52%

The important distinction is that missing future production values are not a data-quality failure; they are the values the model is supposed to forecast. Weather variables were complete in this run, while some forecast-day fields are missing because they depend on future or lagged production values.

6 Exploratory Analysis

Figure 1 shows the last 365 observed days of daily solar production. The series has strong seasonality and clear day-to-day variation, which motivates combining lagged production with weather regressors.

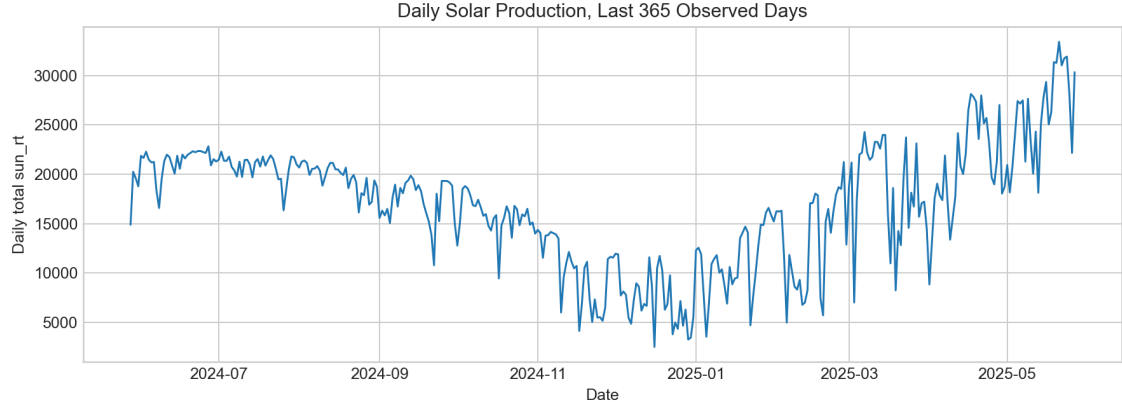


Figure 1: Daily solar production over the last 365 observed days.

Table 4 lists the strongest correlations with daily production among engineered daily variables. The 3-day lag is the strongest single predictor, while radiation-related variables are also positively associated with production. Cloud cover is negatively correlated, as expected.

Table 4: Top correlations with daily solar production.

Variable	Correlation with daily production
3-day lagged production sum	0.903
Maximum shortwave radiation	0.430
Mean shortwave radiation	0.420
Effective radiation	0.353
Mean cloud cover	-0.312
Mean temperature	0.311

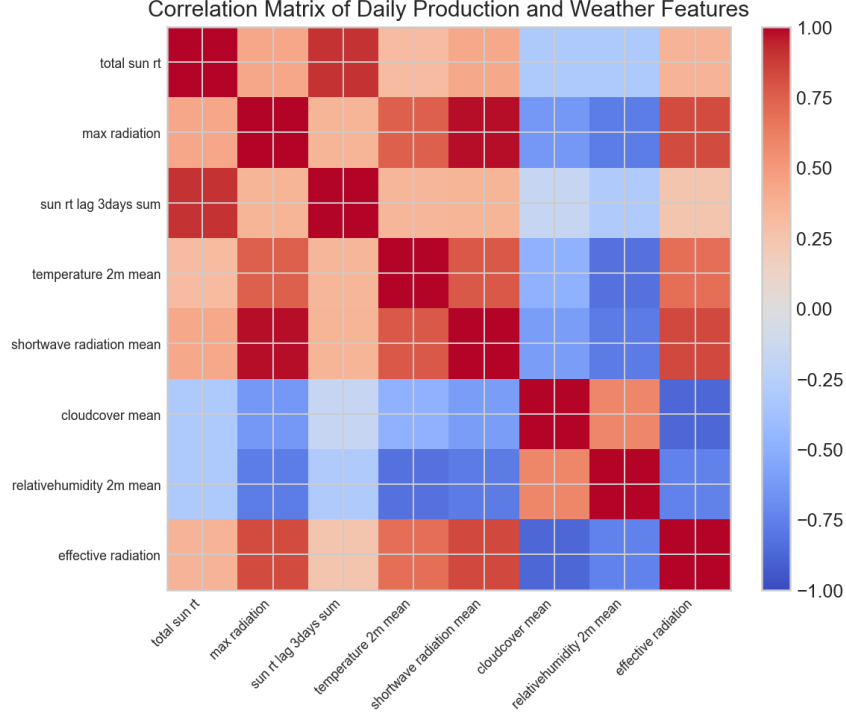


Figure 2: Correlation matrix for daily production and weather-derived variables.

7 Modeling Approach

7.1 Daily ARIMAX-Style Model

The final model uses SARIMAX as an ARIMAX-style model. The endogenous variable is log-transformed total daily production. The exogenous regressors are the engineered weather variables, the lagged production feature, day-of-week dummies, and a trend index. The model order is:

$$(p, d, q) = (1, 1, 1).$$

For a target day $d + 1$, training observations are restricted to dates up to $d - 2$. This design avoids using production observations that would not be known at forecast time.

7.2 Hourly Distribution

The SARIMAX model forecasts total daily production. The required output, however, is hourly. To convert the daily forecast into 24 values, the pipeline learns historical hourly production ratios:

$$r_h = \frac{\text{production at hour } h}{\text{daily production total}}.$$

The ratios are grouped by year, ISO week, and hour. For a forecast date, the matching weekly profile is used when available. If not available, the code falls back to the previous week, then to the same ISO week averaged across years, and finally to equal 24-hour allocation if no historical profile exists.

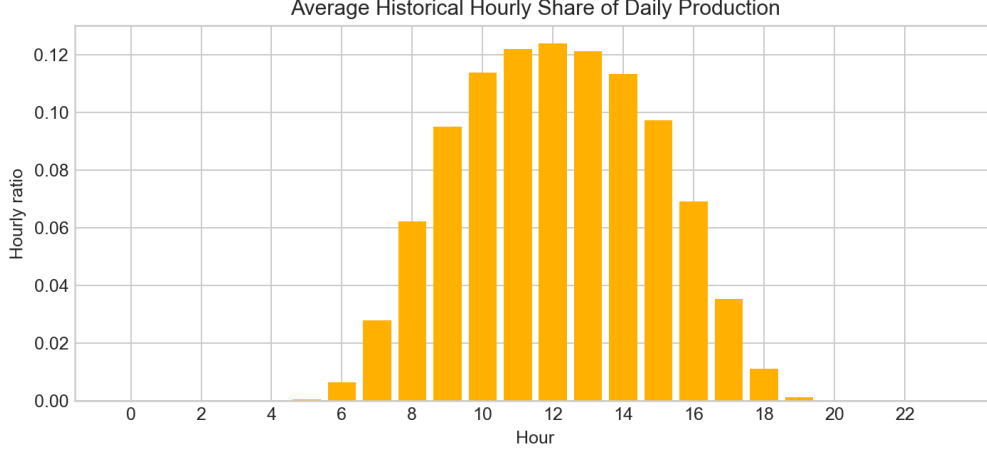


Figure 3: Average historical hourly share of daily production.

8 Validation Design

Because the current hosted production data ends on 2025-05-27, live forecasts for 2026-06-01 cannot be scored against actual production. To obtain a local validation estimate, the pipeline was rerun with the forecast date fixed at 2025-05-27. The last 10 forecasted daily totals from that anchored run were then compared with actual production values.

The main evaluation metric is WMAPE:

$$\text{WMAPE} = \frac{\sum_t |y_t - \hat{y}_t|}{\sum_t y_t} \times 100.$$

MAE and RMSE are also reported for scale-aware interpretation.

9 Results

Table 5: Backtest metrics around the last available production date.

Level	Observations	MAE	RMSE	WMAPE
Daily total forecast	10 days	2,214.96	2,566.09	7.44%
Hourly forecast	240 hours	115.05	204.04	9.28%

Figure 4 compares the daily forecasts with actual daily production in the recent validation window. The model follows the general level of the production series, while individual daily errors remain due to weather variation and the two-stage daily-to-hourly structure.

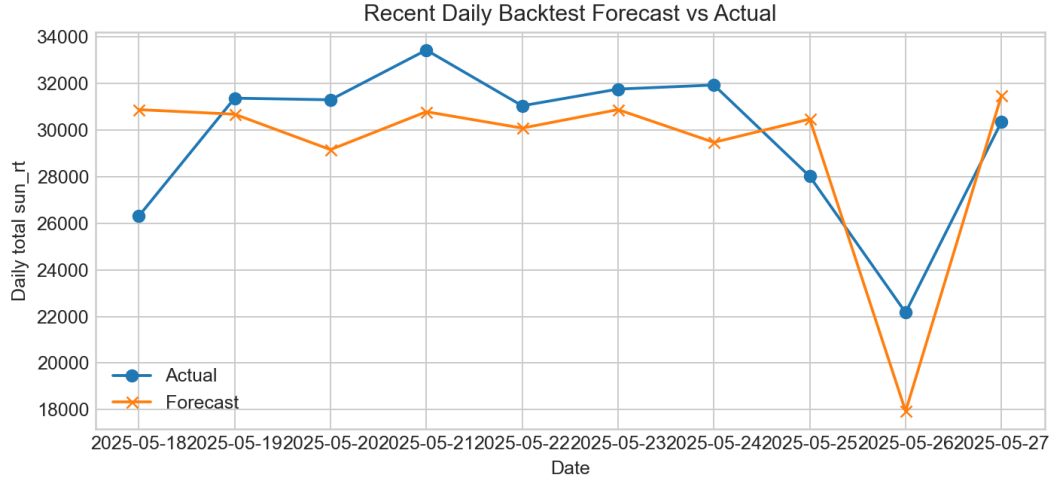


Figure 4: Recent daily backtest forecast versus actual production.

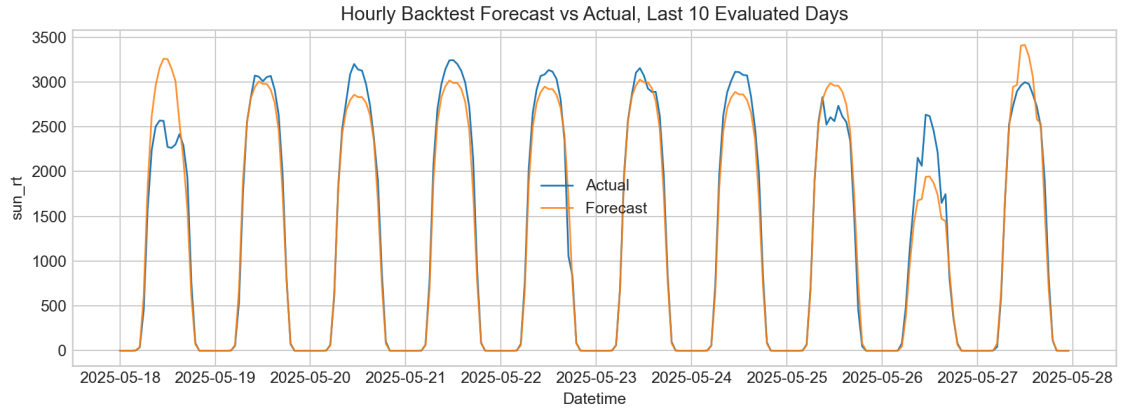


Figure 5: Hourly backtest forecast versus actual production for the evaluated period.

10 Operational Pipeline

Table 6 summarizes the final implementation flow. The cleaned code follows the same logical structure, with separate functions for data loading, feature construction, daily forecasting, and hourly allocation.

Table 6: Operational flow of `forecast.py`.

Step	Main action	Reliability note
Load data	Fetch production and Open-Meteo weather data.	Weather and production are merged by timestamp.
Build features	Create lagged production, weather aggregates, effective radiation, trend, and week-day variables.	Uses lagged production to avoid same-day leakage.
Train daily model	Fit SARIMAX with exogenous regressors.	Model is retrained at runtime.
Forecast daily total	Predict future daily total on log scale and transform back.	Negative predictions are clipped to zero.
Distribute hourly	Apply historical hourly production ratios.	Falls back to previous week or same-week profile across years.
Print output	Return 24 numeric hourly forecasts.	Compatible with the project submission format.

11 Discussion

11.1 Strengths

The final approach is interpretable and operationally practical. It uses domain signals that are directly related to solar generation: shortwave radiation, cloud cover, temperature, and historical production. The two-stage design also reduces the burden on the SARIMAX model by forecasting daily totals instead of modeling all hourly noise directly.

11.2 Limitations

The main limitation is data availability. The hosted production data currently ends on 2025-05-27, so current 2026 forecasts cannot be scored unless updated actual production is provided. The model also uses a fixed SARIMAX order rather than automatically tuning (p, d, q) . Finally, the hourly allocation step is heuristic: it uses historical ratios rather than a separate hourly regression model.

11.3 Possible Improvements

The project could be improved by adding a reproducible local dataset, a dedicated rolling back-test script, automatic SARIMAX parameter selection, and alternative models such as regularized regression, random forests, or gradient boosting. It would also be useful to refactor `forecast_helper.py` to remove duplicated weather API logic and to add command-line arguments for forecast date and output path.

12 Conclusion

This project implements a complete next-day solar production forecasting pipeline for the IE360 Time Series Analysis term project. The final model combines historical production, weather forecasts, engineered calendar features, and an ARIMAX-style SARIMAX model. The local

backtest around the latest observed production period produced 7.44% daily WMAPE and 9.28% hourly WMAPE. The resulting implementation is compact, reproducible at runtime, and aligned with the required 24-hour forecast output format.

References

- IE360 Time Series Analysis term project materials, Boğaziçi University.
- Open-Meteo Weather API: <https://open-meteo.com/>.
- Hyndman, R. J. and Athanasopoulos, G. *Forecasting: Principles and Practice*.
- Durbin, J. and Koopman, S. J. *Time Series Analysis by State Space Methods*.
- Statsmodels SARIMAX documentation: <https://www.statsmodels.org/>.